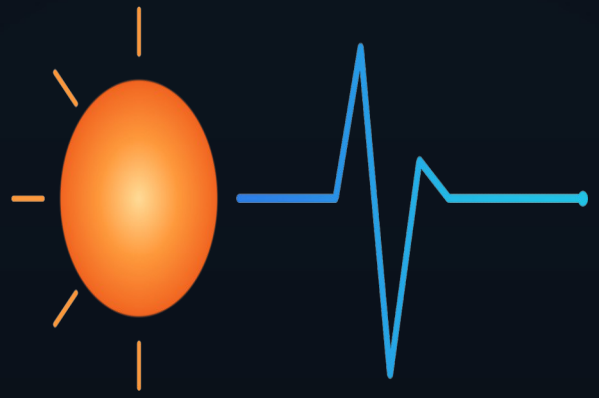


SOHelioraX

AI-Powered Solar Flare Nowcasting & Risk Forecasting System

Built on Real ISRO Aditya-L1 Data · SoLEXS + HEL1OS

Bharatiya Antariksh Hackathon 2026

Archita Tripathi · Manthan Dhanraj | Sharda University, Greater Noida

The Problem

Why real-time solar-flare monitoring matters

Disruptive Impact

Solar flares disrupt satellites, GPS, aviation, power grids and communication networks within minutes of eruption.

Foreign Dependence

Most operational tools rely on NOAA / GOES data and fixed thresholds — not on India's own solar mission.

Untapped Aditya-L1 Data

Level-1 data is mostly used for post-event study; there is no easy real-time, human-readable warning system.

Our Solution

Raw Aditya-L1 telemetry → instant, human-readable intelligence

01 Reads

Live SoLEXS & HEL1OS Level-1 FITS files

02 Extracts

Statistical X-ray light-curve features

03 Classifies

Flares A–X via Random Forest + physics

04 Scores

4-level risk (Green / Yellow / Orange / Red)

05 Forecasts

1H / 2H / 4H probability + confidence

06 Alerts

Email / SMS warning with ~30 min lead

Key Features

Live Data Pipeline

Auto-reads SoLEXS & HEL1OS Level-1 FITS with feature extraction.

Nowcasting Engine

Physics + ML hybrid — Random Forest classifies flares in real time.

Forecasting Panel

1H / 2H / 4H probabilistic forecasts with confidence scores.

Dual Risk Gauges

Live risk gauge + a 739-day historical risk baseline.

Lead-Time Alerts

~30-min early warning before peak, with email / SMS alerts.

Interactive Explorer

Browse all 739 days on a month-wise calendar with analytics.

System Architecture

1 Data Layer

Aditya-L1 FITS · SoLEXS Level-1 · HEL1OS Level-1 (ISSDC PRADAN)

2 Processing Layer

Ingestion → Preprocessing → Feature Engineering → Structured Dataset

3 AI / Logic Layer

Peak Detection → Random Forest Classifier → Risk Forecaster (ML)

4 Presentation Layer

Flask Web App · Interactive Dashboard · Email / SMS Alerts

Dataset — Real Aditya-L1 Data

739

DAYS OF REAL OBSERVATIONAL DATA

Instruments

SoLEXS (Soft X-ray) + HEL1OS (Hard X-ray)

Source

ISRO Aditya-L1 · ISSDC PRADAN portal

Features / day

peak, mean, std, median, p95 (×2 sensors)

Labeling

SoLEXS soft X-ray peak (GOES standard)

Flare Classification Thresholds

SoLEXS soft X-ray peak counts

Class	Threshold	Level
A	< 150	Quiet
B	< 750	Minor
C	< 3,500	Moderate
M	< 18,000	Strong
X	≥ 18,000	Extreme

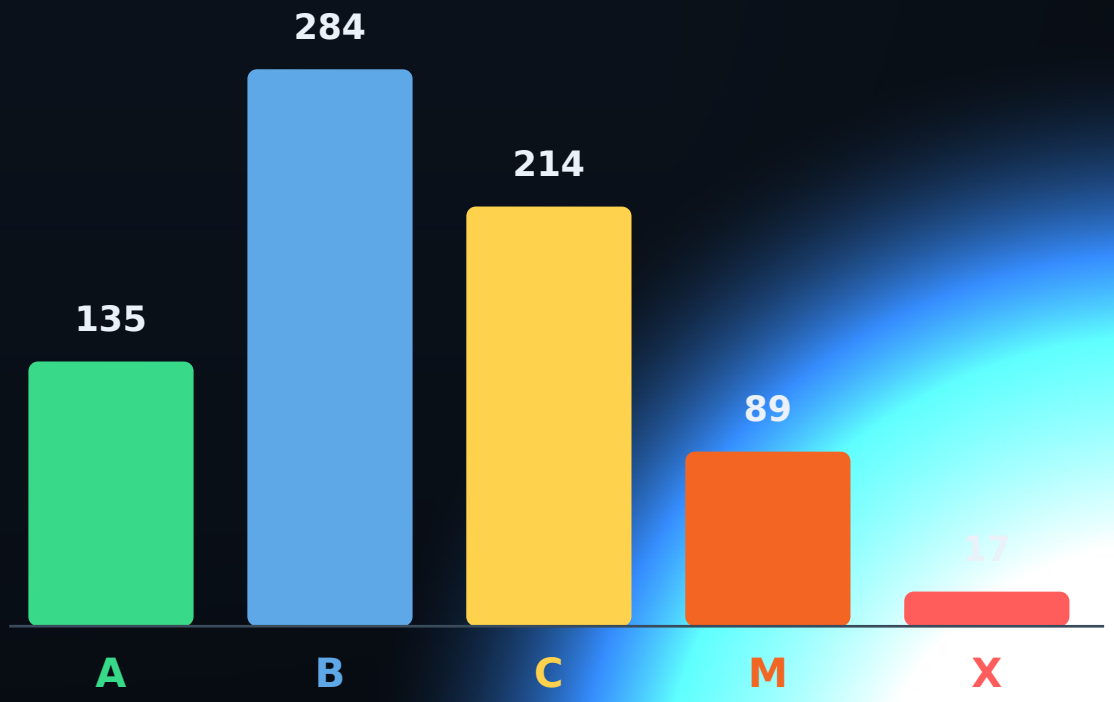
Machine Learning Model

97.7% ± 1.1%

CROSS-VALIDATED ACCURACY (5-fold stratified)

- Random Forest Classifier — 300 trees, balanced classes
- Honest 5-fold Stratified Cross-Validation (mean ± std)
- Hybrid: ML probabilities blended 50/50 with physics
- Outputs: flare class, C/M/X %, confidence, risk score

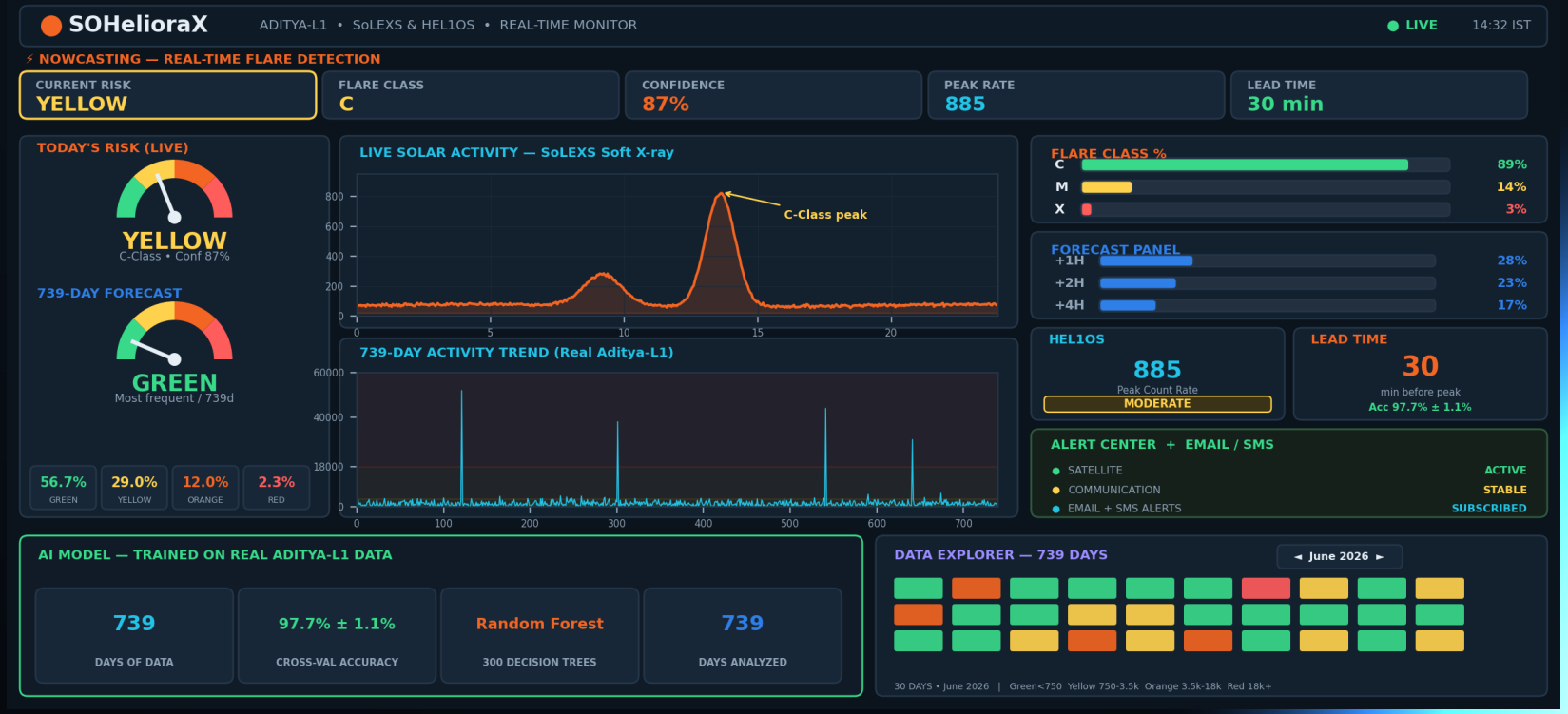
Flare-Class Distribution (739 days)



Realistic distribution — X-class rare, B/C common

Interactive Dashboard & Results

97.7%	739	~30 min	4-Level
CV Accuracy	Days of Data	Lead Time	Risk Scoring



Live nowcasting · dual gauges · 739-day explorer · flare-class % · forecasts · email/SMS alerts

Tech Stack & Deployment

Backend & ML

- Python · Flask
- scikit-learn (Random Forest)
- pandas · NumPy
- Astropy (FITS) · joblib

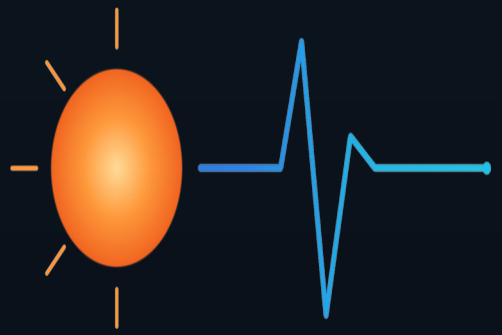
Frontend

- HTML5 · CSS3 · JavaScript
- Dark dashboard UI
- Custom SVG gauges
- Responsive panels

Deployment

- Gunicorn (WSGI)
- Render (cloud hosting)
- gunicorn app:app
- Lightweight & reproducible

Deploy in 3 steps: **1** Push repo to GitHub **2** Render → New Web Service (root: Backend) **3** Start: gunicorn app:app



Thank You

SOHelioraX — Real-time Space Weather Intelligence

Accurate Nowcasting · Early Warnings · Operational Readiness

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